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Tackling Global Issues: Change

Rebecca Killick(r.killick@lancs.ac.uk)
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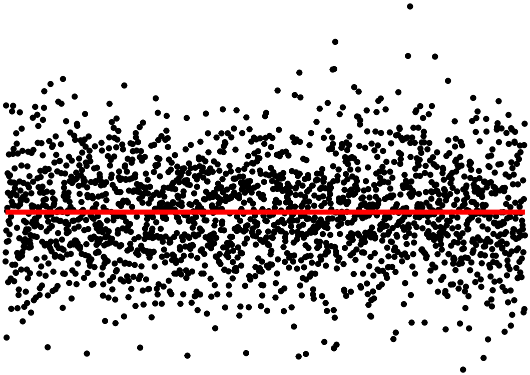
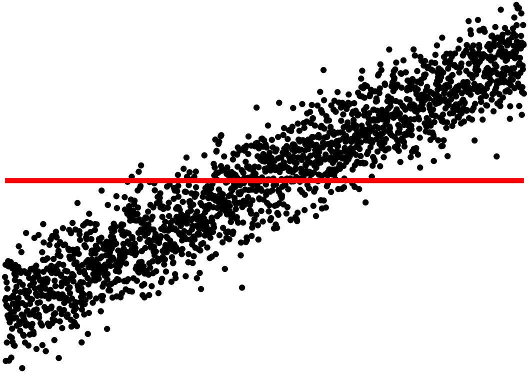
Model development



Model development



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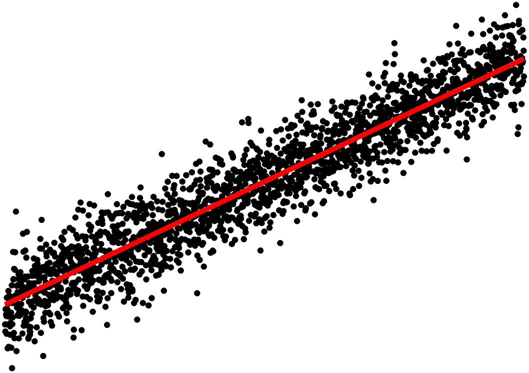
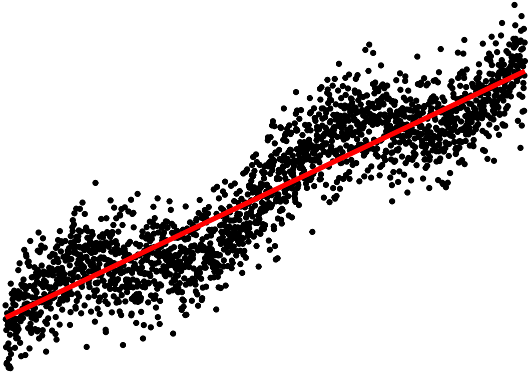
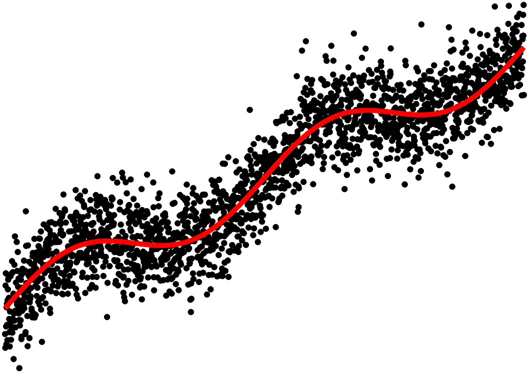


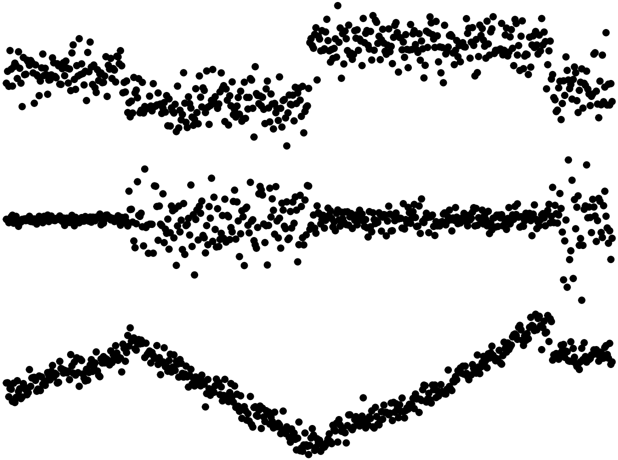
Model development



Model development



Model Development



Outline

- What are changepoints?
- Tour of changepoint problems and applications
  - Speed & accuracy
  - Autocorrelation
  - Model choice
  - Multivariate
- Open problems
  - Autocorrelation
  - Multivariate
  - Sensitivity to data
  - Online inference

What are changepoints?

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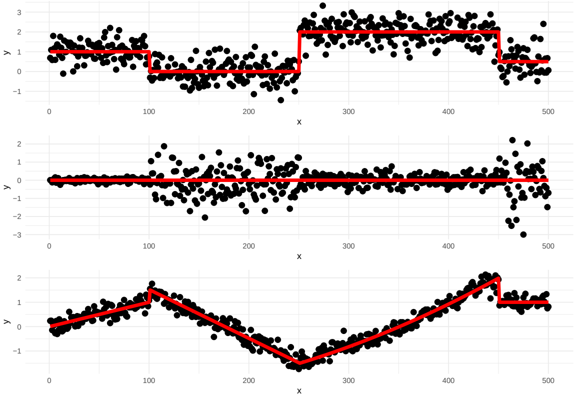
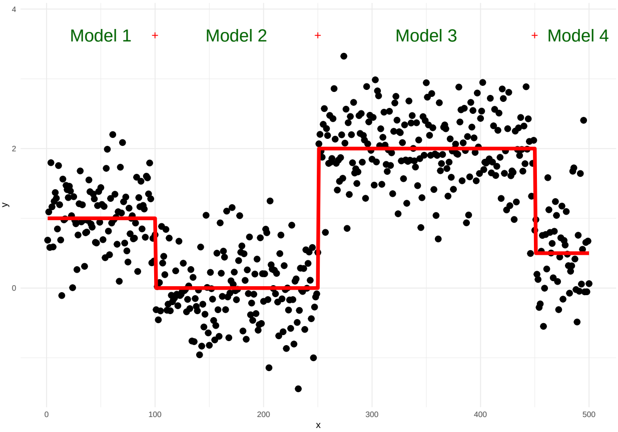
For data $y_1, \ldots, y_n$, if a changepoint exists at τ, then $y_1, \ldots, y_\tau$ differ from $y_{\tau+1}, \ldots, y_n$ in some way.

There are many different types of change.



Problem

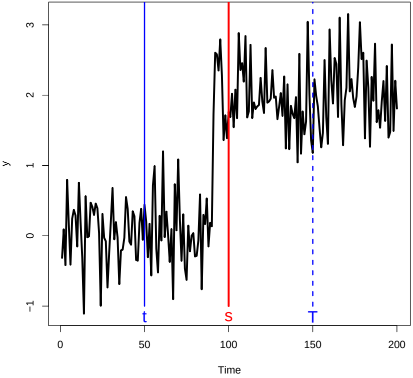


- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

Speed / Accuracy

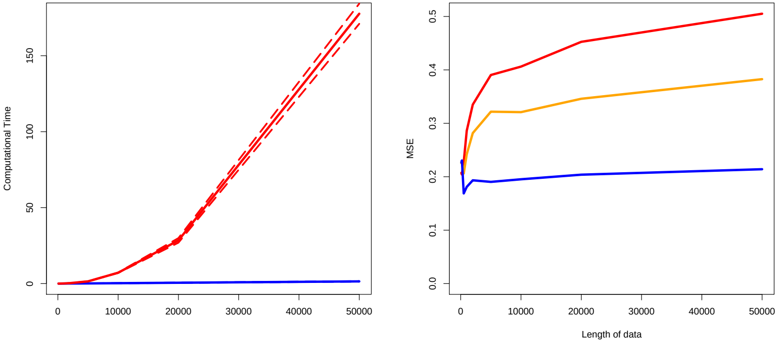
Computational complexity in 2010: $\mathcal{O}(n^2)$ exact, $\mathcal{O}(nlogn)$ approx.

$\mathcal{O}(n^2)$ solution book keeps the last changepoint location. BUT ...



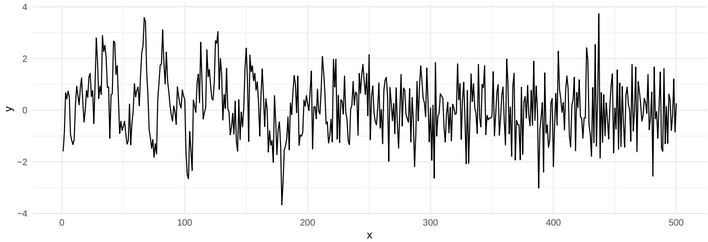
Speed / Accuracy

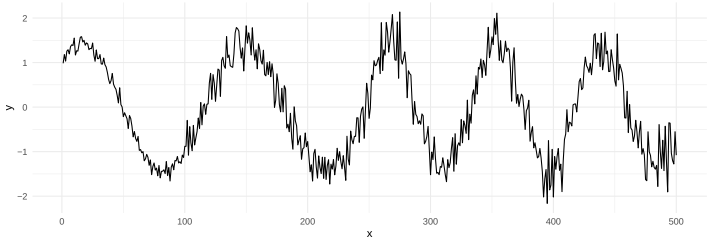
. . . resulting in an $\mathcal{O}(n)$ algorithm if changepoints occur regularly.



SPEED + ACCURACY = PELT!

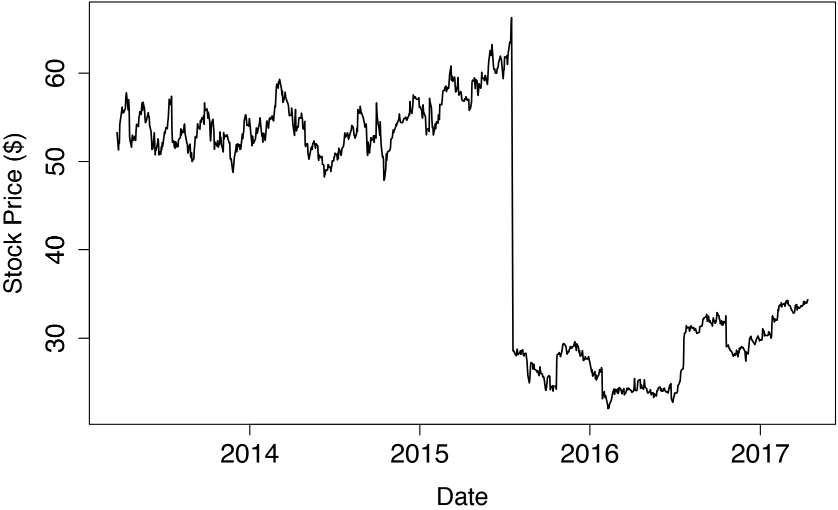
More complicated changes





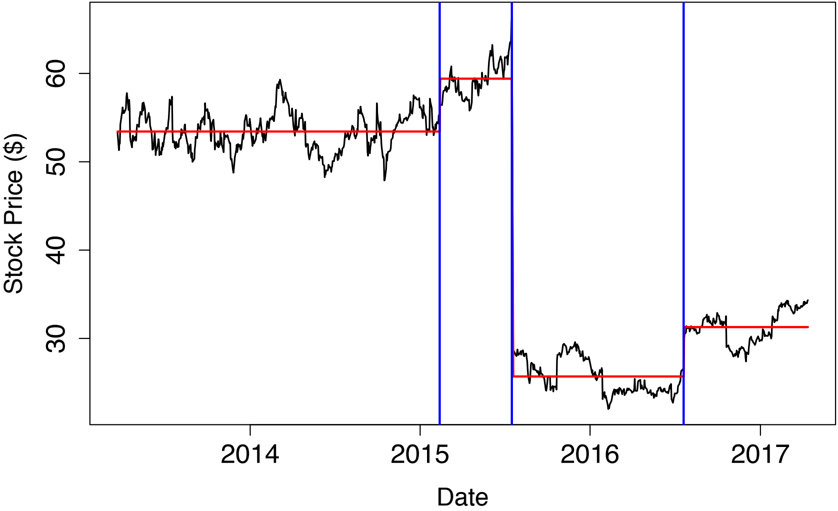
Stock Data

Changes are hard to identify with background correlation

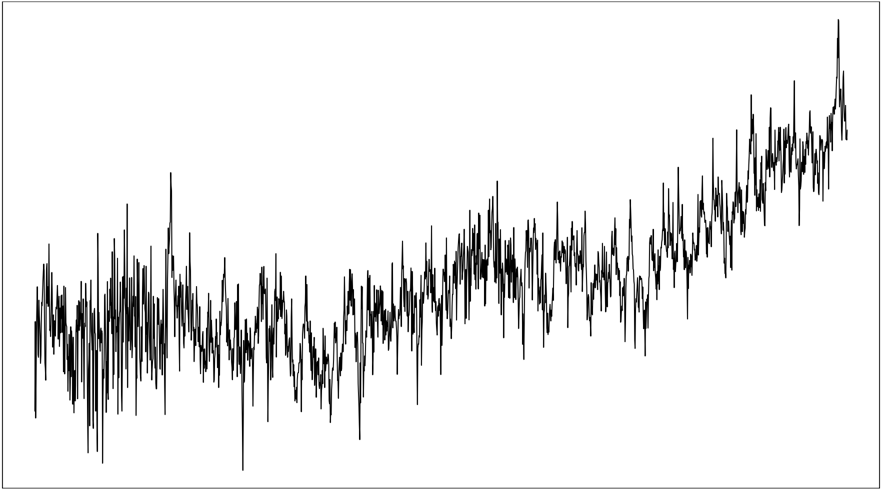


Stock Data

Change in mean with non-stationary errors

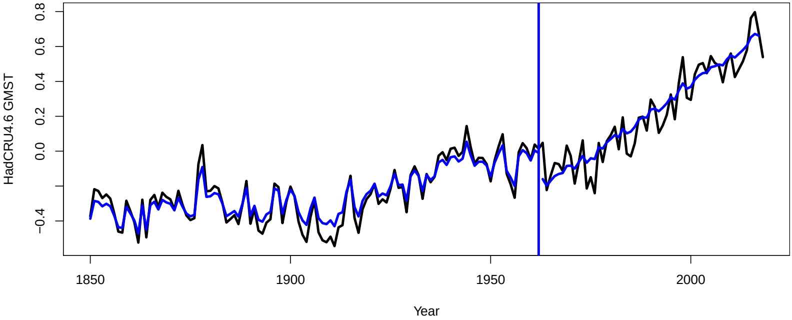


Natural correlation



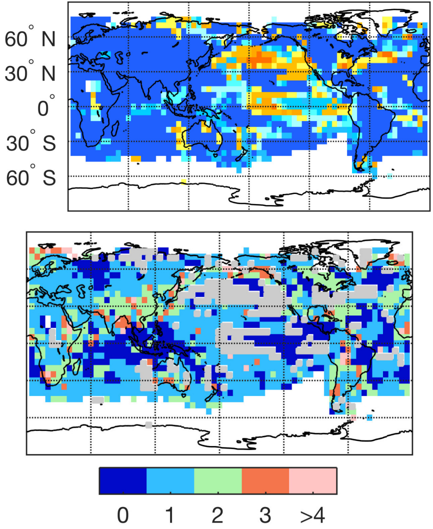
Climate Change

- Has there been a recent hiatus in global warming?
- Analysed several surface temperature time series estimates.



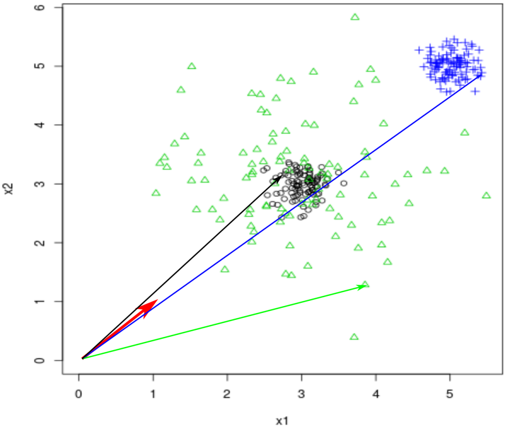
- Developed automatic model selection including trend, autocorrelation and changepoint detection
- This indicates no hiatus in all series.

Climate Dynamics



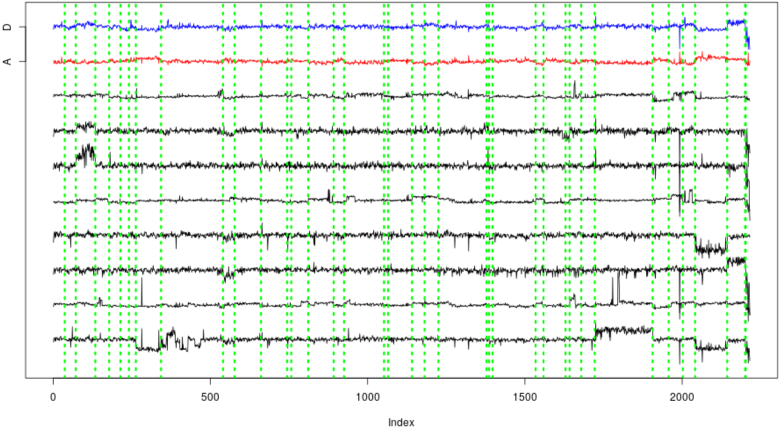
Health

- Which genomic markers could be indications for disease?
- Mean and variance changes



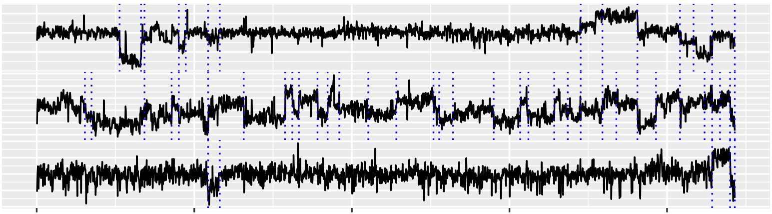
Health

- Which genomic markers could be indications for disease?
- Use two transformations instead of one



Health

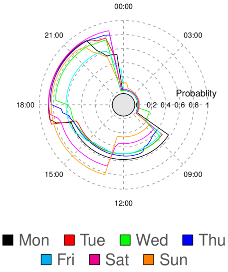
- Which genomic markers could be indications for disease?
- Only markers present in lots of patients are worth investigating.

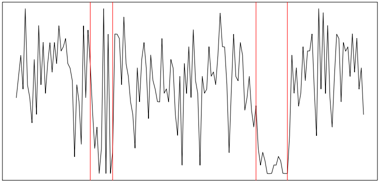


- Developed computationally efficient subset multivariate changepoint methodology.
- Only 10 changes are present in more than 60% of the patients.
- Over 50 detected by competitors.

OP: Aging

- Passively monitor patients at home
- Desire to have early intervention before crisis point is reached

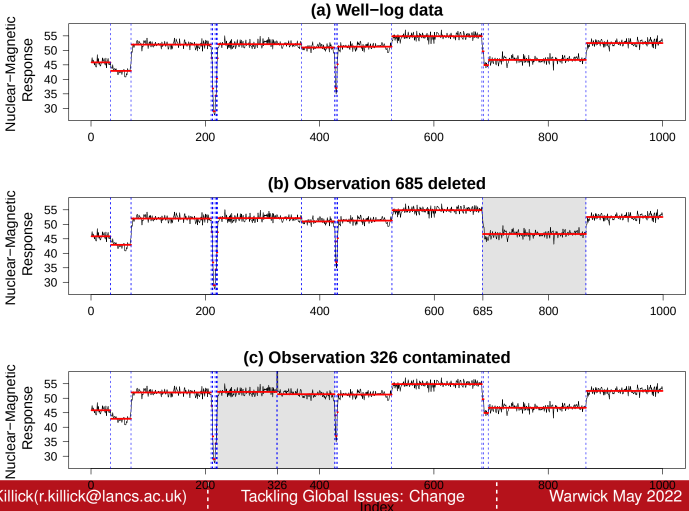




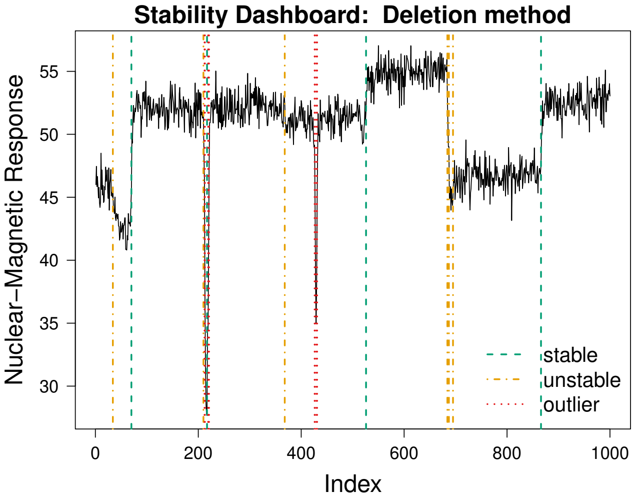
- View data as circular, e.g., each day.
- Developed circular changepoint detection to detect changes in activity levels within a day
- Changes in the detected routine signal early intervention

OP: Sensitivity

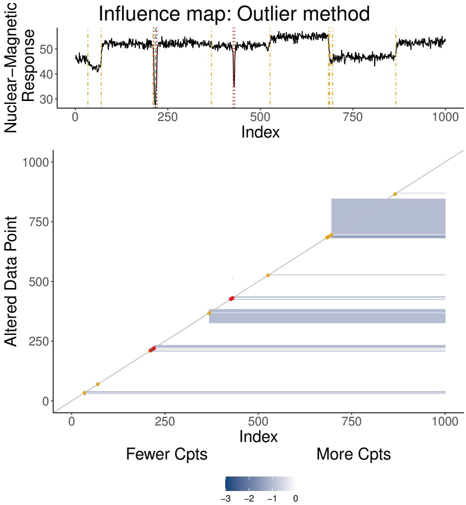
- Changepoint locations are sensitive to the observed data
- Some changepoint algorithms are more sensitive than others.



OP:Sensitivity



OP:Sensitivity



Open Problems

- Autocorrelation
- Multivariate data
- Model complexity
- Online changes

Summary

- Changepoints occur in most data
- Detecting and documenting them improves analyses
- Multivariate, online, big data mostly isn't a problem
- But there are still plenty of open problems in the field
- I enjoy working with messy real life data ...
- ... as often it sparks my next research challenge.

Papers discussed

Preprints of all available at: www.lancs.ac.uk/~killick/pub.html

PELT: https://doi.org/10.1080/01621459.2012.737745

LMvsCpts: https://doi.org/10.1007/s11222-017-9731-0 & https://doi.org/10.1002/env.2568

Model Choice: https://doi.org/10.1175/JCLI-D-17-0863.1 & https://doi.org/10.1002/qre.2712

Multivariate: https://doi.org/10.1038/s41598-019-46836-y & https://doi.org/10.1007/s11222-020-09940-y

Autocorrelation: https://doi.org/10.1002/sta4.351

Recent papers

Circular time: http://doi.org/10.1111/rssc.12472

Sensitivity (accepted in JCGS): https://arxiv.org/abs/2107.10572

Multivariate: https://arxiv.org/abs/2108.07340

Autocorrelation: https://arxiv.org/abs/2102.10669

Online: Email for preprint, code:
https://github.com/grundy95/changepoint.forecast

Autocorrelation Change

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Change in variance with nonstationary errors:
https://doi.org/10.1002/env.2576

Changes in second order structure within nonstationary series:
http://dx.doi.org/10.1214/13-EJS799

Uncertainty of autocovariance changes:
https://doi.org/10.1080/00401706.2014.902776